Sheth L.U.J & Sir M.V College of Science
Subject :- Artificial Intelligence
Practical no 5

Aim :- K-Nearest Neighbors (K-NN)

- **Implement the K-NN algorithm for classification or regression.**
- **Apply the K-NN algorithm to a given dataset and predict the class or value for test data.**
- **Evaluate the accuracy or error of the predictions and analyze the results.**

Description :-

No.	Function / Code	Description
1	`pd.read_csv()`	Loads the Kaggle Red Wine Quality dataset.
2	`X` and `y`	Separates the wine features and the `quality` target variable.
3	`train_test_split()`	Divides the dataset into 80% training and 20% testing data.
4	`StandardScaler()`	Standardizes the feature values so that all features have a similar scale.
5	`KNeighborsClassifier()`	Creates the K-Nearest Neighbors classification model using K = 5.
6	`fit()`	Trains the K-NN model using the training data.
7	`predict()`	Predicts the wine quality classes for the test data.
8	`accuracy_score()`	Calculates the accuracy of the K-NN model.
9	`1 - accuracy`	Calculates the prediction error of the model.

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10	`classification_report()`	Displays precision, recall, and F1-score for each class.
11	`confusion_matrix()`	Creates a matrix showing the correct and incorrect classifications.
12	`plt.imshow()`	Displays the confusion matrix graphically.
13	`KNeighborsClassifier()` in loop	Tests different K values from 1 to 15.
14	`accuracy_score()` in loop	Calculates accuracy for each K value.
15	`plt.plot()`	Plots K value vs. accuracy to compare model performance.
16	`max()` and `index()`	Finds the K value that produces the highest accuracy.

Code :-

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```python
import pandas as pd
import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import accuracy_score
from sklearn.metrics import confusion_matrix
from sklearn.metrics import classification_report


data = pd.read_csv("winequality-red.csv")

print("Dataset loaded successfully!")

print("\nFirst 5 rows:")
print(data.head())

print("\nDataset shape:")
print(data.shape)

print("\nColumn names:")
print(data.columns.tolist())


X = data.drop("quality", axis=1)

y = data["quality"]
```

```python
y = data["quality"]


print("\nNumber of features:", X.shape[1])

print("Number of samples:", X.shape[0])

print("\nTarget classes:")
print(sorted(y.unique()))


X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.20,
    random_state=42
)


print("\nTraining samples:", X_train.shape[0])

print("Testing samples:", X_test.shape[0])


scaler = StandardScaler()

X_train_scaled = scaler.fit_transform(X_train)

X_test_scaled = scaler.transform(X_test)
```


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```python
print(y_pred[:15])


accuracy = accuracy_score(
    y_test,
    y_pred
)

error = 1 - accuracy


print("\n===================================")
print("K-NN MODEL PERFORMANCE")
print("===================================")

print(
    "Accuracy:",
    round(accuracy * 100, 2),
    "%"
)

print(
    "Error:",
    round(error * 100, 2),
    "%"
)


print("\nClassification Report:")
```

```python
print(
    classification_report(
        y_test,
        y_pred,
        zero_division=0
    )
)


cm = confusion_matrix(
    y_test,
    y_pred
)


print("\nConfusion Matrix:")

print(cm)


classes = sorted(y.unique())

plt.figure(figsize=(8, 6))

plt.imshow(
    cm,
    interpolation="nearest"
)
```


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```
plt.title("K-NN Confusion Matrix")

plt.xlabel("Predicted Quality")

plt.ylabel("Actual Quality")

plt.colorbar()


plt.xticks(
    range(len(classes)),
    classes
)

plt.yticks(
    range(len(classes)),
    classes
)


for i in range(len(classes)):

    for j in range(len(classes)):

        plt.text(
            j,
            i,
            cm[i, j],
            ha="center",
            va="center"
```

```


plt.tight_layout()

plt.show()


k_values = range(1, 16)

accuracy_values = []


for k_value in k_values:

    model = KNeighborsClassifier(
        n_neighbors=k_value
    )

    model.fit(
        X_train_scaled,
        y_train
    )

    predictions = model.predict(
        X_test_scaled
    )

    score = accuracy_score(
        y_test,
        predictions
```


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```python
    accuracy_values.append(score)


best_index = accuracy_values.index(
    max(accuracy_values)
)

best_k = list(k_values)[best_index]

best_accuracy = accuracy_values[best_index]


print("\n====================================")
print("BEST K VALUE")
print("====================================")

print("Best K:", best_k)

print(
    "Best Accuracy:",
    round(best_accuracy * 100, 2),
    "%"
)


plt.figure(figsize=(8, 5))

plt.plot(
    k_values,
```

```python
    marker="o"
)

plt.xlabel("K Value")

plt.ylabel("Accuracy")

plt.title("K-NN: K Value vs Accuracy")

plt.xticks(
    list(k_values)
)

plt.grid(True)

plt.tight_layout()

plt.show()
```

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OutPut :-

```
Python 3.14.6 (tags/v3.14.6:c63aec6, Jun 10 2026, 10:26:10) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
======== RESTART: C:/Users/Vinod Mali/Desktop/Ai/T092 practical no 8.py ========
Dataset loaded successfully!

First 5 rows:
   fixed acidity  volatile acidity  citric acid  ...  sulphates  alcohol  quality
0            7.4              0.70         0.00  ...       0.56      9.4        5
1            7.8              0.88         0.00  ...       0.68      9.8        5
2            7.8              0.76         0.04  ...       0.65      9.8        5
3           11.2              0.28         0.56  ...       0.58      9.8        6
4            7.4              0.70         0.00  ...       0.56      9.4        5

[5 rows x 12 columns]

Dataset shape:
(1599, 12)

Column names:
['fixed acidity', 'volatile acidity', 'citric acid', 'residual sugar', 'chlorides', 'free sulfur dioxide', 'total sulfur dioxide', 'density', 'pH', 'sulphates', '
alcohol', 'quality']

Number of features: 11
Number of samples: 1599

Target classes:
[np.int64(3), np.int64(4), np.int64(5), np.int64(6), np.int64(7), np.int64(8)]

Training samples: 1279
```

```
Testing samples: 320

K-NN model trained successfully!
K value: 5

First 15 Actual Values:
[6 5 6 5 6 5 5 5 5 6 7 3 5 5 6]

First 15 Predicted Values:
[5 5 6 6 6 5 5 5 6 6 7 6 6 5 5]

==================================
K-NN MODEL PERFORMANCE
==================================
Accuracy: 54.69 %
Error: 45.31 %

Classification Report:
              precision    recall  f1-score   support

           3       0.00      0.00      0.00         1
           4       0.00      0.00      0.00        10
           5       0.59      0.68      0.63       130
           6       0.51      0.55      0.53       132
           7       0.54      0.36      0.43        42
           8       0.00      0.00      0.00         5

    accuracy                           0.55       320
   macro avg       0.27      0.26      0.26       320
weighted avg       0.52      0.55      0.53       320
```


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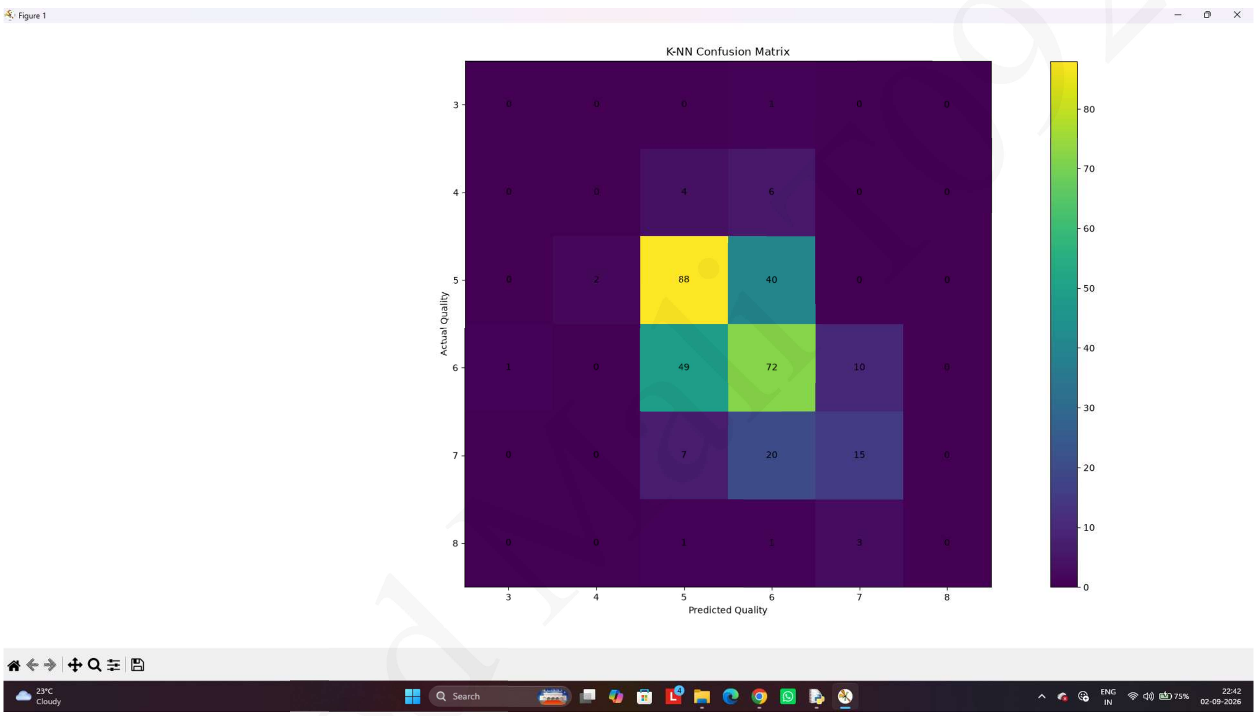
```
Confusion Matrix:
[[ 0  0  0  1  0  0]
 [ 0  0  4  6  0  0]
 [ 0  2 88 40  0  0]
 [ 1  0 49 72 10  0]
 [ 0  0  7 20 15  0]
 [ 0  0  1  1  3  0]]

======== RESTART: C:/Users/Vinod Mali/Desktop/Ai/T092 practical no 8.py ========
```

Graphs :-



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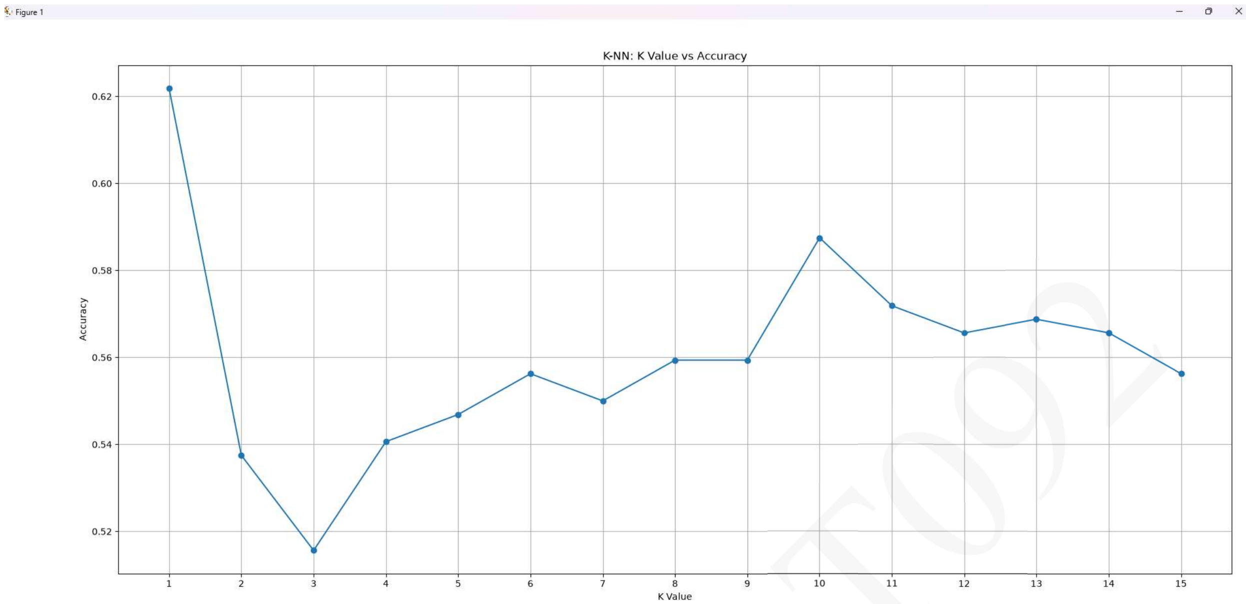

Observations :-

1. Confusion Matrix

The confusion matrix shows that the K-NN model correctly classifies most wine-quality samples, while a smaller number of samples are misclassified between neighboring quality classes. Higher values along the diagonal indicate better classification performance.

2. K Value vs Accuracy Graph

The graph shows that the accuracy changes with different values of K. Very small or large K values may give lower accuracy, while the best K value provides the highest accuracy. This shows that selecting an appropriate K is important for achieving better K-NN performance.

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